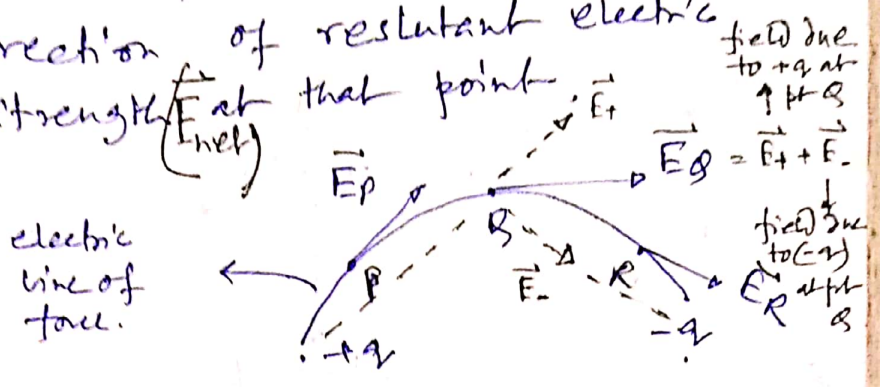
Electric lines of force.
or Electric field lines.

Electric lines of force are imaginary lines or curves which are helpful to visualize an electric field.

Definition: Electric line of force is an imaginary curve such that the tangent at any point of it gives the direction of resultant electric field strength ($\vec{E}_{net}$) at that point.

(Board exam)



It has following properties:

(1) Tangent to line of force at a point gives direction of net Electric field.

(2) It emerge from a positive charge and terminate on negative charge.

(3) number of lines of force per unit crossectional area in a region (ie density of field lines) is proportional to intensity of electric field (E)

JEE

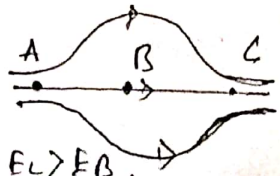
(closeness)

ex:

$|\vec{E}_A| > |\vec{E}_B|$

(4) electric line of force never form closed loops.

ex:



$E_A = E_C$

But $E_A \sim E_C > E_B$

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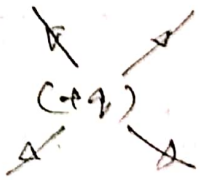
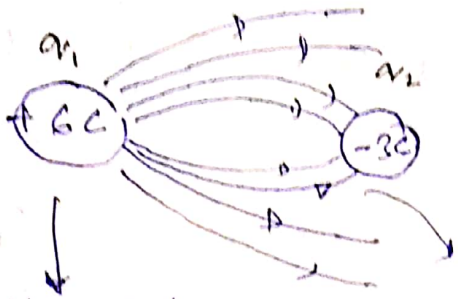
5. Number of lines of force leaving a +ve charge or entering a -ve charge is proportional to magnitude of the charge.

JEE

For example: If 100 lines of force emerge from a +30 Coulomb charge, then 50 lines of force will emerge from +15C charge or terminate to (-15C) charge.



ex:



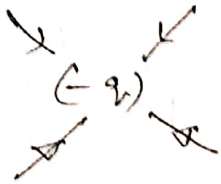
$\left|\frac{q_1}{q_2}\right| = \frac{6}{3} = 2$

but $\frac{q_1}{q_2} = -2$ [$\because q_1, q_2$ are of opposign]

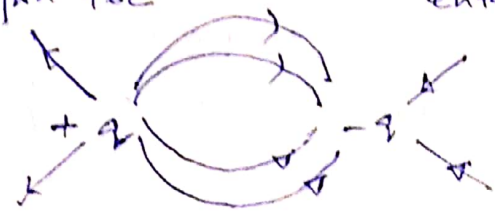
4 lines of force have emerged from +6C

2 lines of force have entered (-3C).

ex!



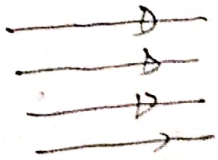
ex!



Board! 6) exam Two lines of never intersect. If it happens then two tangents can be drawn at their point of intersection, i.e. intensity E at that pt will have two directions which is absurd.

7) (a) For a uniform electric field (E = constant) lines of force are straight, parallel, lines with equal spacing.

Like



E = constant.

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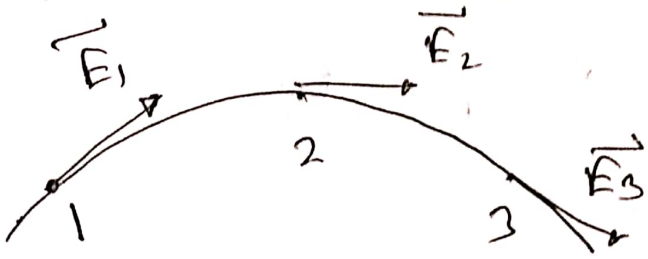
(b) Curved line of force represents non-uniform electric field.

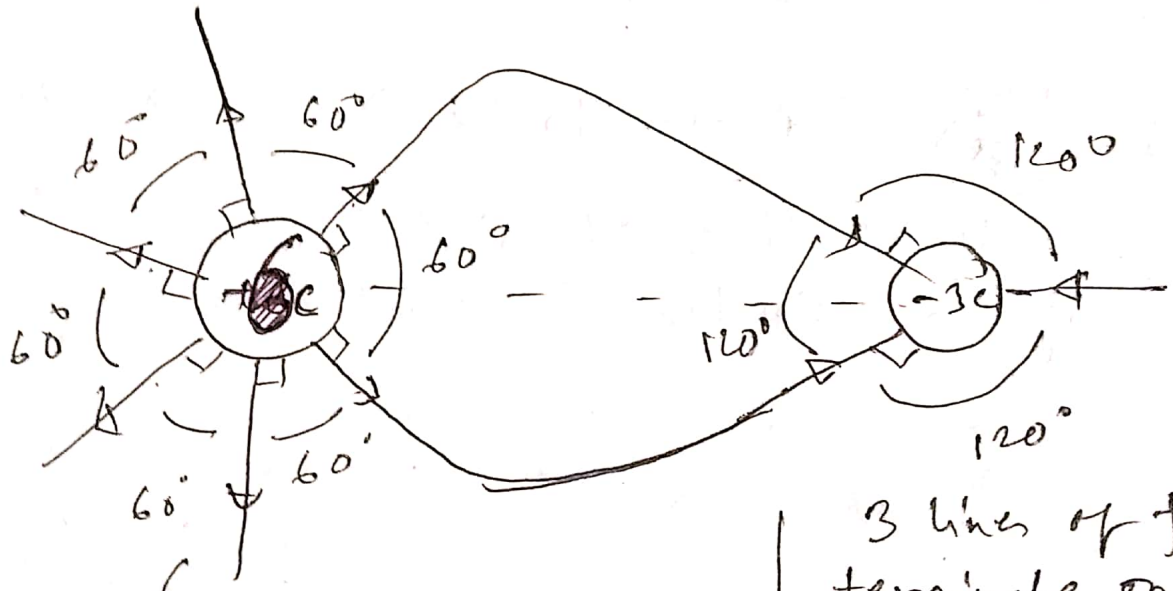


$\vec{E_1} \neq \vec{E_2} \neq \vec{E_3}$

8. If angle betn two adjacent electric line of force emerging / entering an electric charge increases, then it indicates that the magnitude of the concerned charge is ~~weaker~~ / smaller.

JEE

Ex:



Ex: 6 lines of force emerge from (+6C) (stronger) angle betn two adj line of force is 60°

3 lines of force terminate on (-3C) (weaker), so angle betn two adj line of force is 120°

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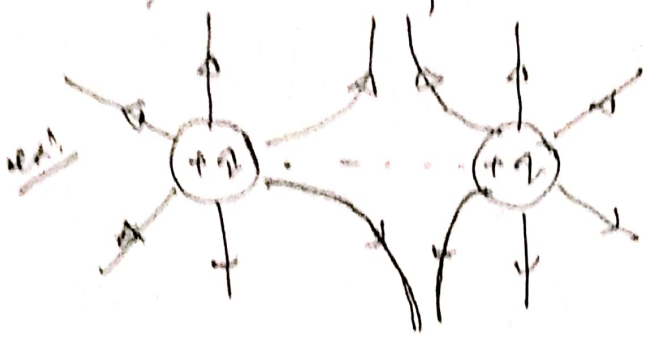
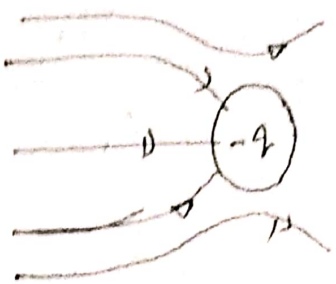
9) Electric lines of force are repelled by +ve charge & attracted by -ve charge

Ex:

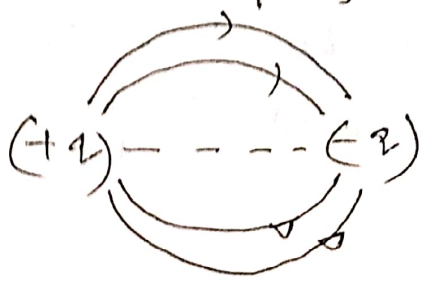


Ex:



(10) Electric lines of force are symmetric (सुडौल) about the line joining two charge.

Ex:



(11) Remember (JEE)

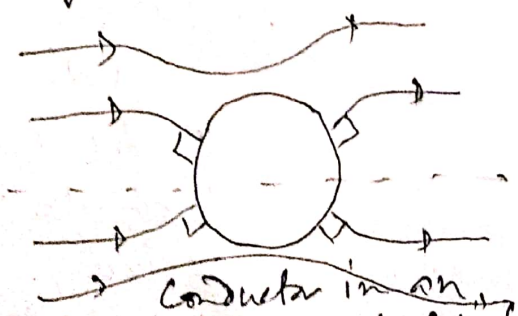
Inside a neutral / charged conductor [on the surface of which charge is at rest] there is no electric field.

(a) So (Reason to be explained later, in following classes)

So there will be no electric line of force inside it.

(b) Remember (JEE) (Reason to be discussed later)

electric line of force is ⊥ to surface of a conductor [neutral / charged]



Conductor in an external electric field.

[in this fig there is no electric line of force inside conductor & they are ⊥ to surface of conductor]

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Electric line of force never form closed loop. (WBHS Exam' 17)

<u>Explanation</u>:

Electric line of force indicates direction of net electric field ($\vec{E}$) (net electric force on a unit +ve charge) & it also indicates the line along which a free +ve charge ~~would~~ moves.

So electric force on a +ve charge & displacement of a +ve charge are either in same dir. ($\theta = 0°$) or angle betn them θ is acute [$\theta < 90°$]. ∴ Work done by conservative electric force ($dW_{el} = F_{el}\, dr \cos\theta$ $= qE\, dr \cos\theta$) in a closed loop is always <u>+ve</u> quantity [∵ $\cos\theta$ = +ve]. It contradicts the fact ~~But we know~~ that work done by a conservative force in a closed loop is always zero.

Hence electric line of force never form <u>closed</u> loop.

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